



Alberto Torrejón
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Alberto Torrejón Valenzuela

PhD in Mathematics

Statistics, Operations Research and Data Science

About Me

"And the child grew and became strong, and the grace of God was upon him."
Luke 2:40

Education

2021 - 2025, PhD in Mathematics

Specialization in Statistics and Operations Research.
Institute of Mathematics, University of Seville, Seville.
Supervisors: *Prof. Justo Puerto Albandoz*, *Prof. Miguel A. Pozo Montañó*.

2020 - 2021, Master's Degree in Mathematics

Specialization in Statistics and Operations Research.
University of Seville, Seville.

2015 - 2020, Double Degree in Mathematics and Statistics

University of Seville, Seville.

Experience

Nov 2025 - Present, Teaching and Research Staff

Postdoctoral FPI Contract
Department of Statistics and Operations Research

Sep 2022 - Nov 2025, Teaching and Research Staff in Training

Predocctoral FPI Contract
Department of Statistics and Operations Research

Aug 2021 - Sep 2022, Research Staff

Institute of Mathematics, University of Seville.
Project: "Nuevos resultados sobre los problemas de diseño y optimización en redes complejas: Aplicaciones al diseño de ciudades inteligentes".
Principal Investigator: *Prof. Justo Puerto Albandoz*

May 2021 - Jul 2021, Data Scientist

Internship Contract
Indorama Venture Química, San Roque, Cádiz.

Languages

- Spanish (native)
- English (B2)

IT Skills

- R (advanced)
- Python (advanced)
- Gurobi (advanced)
- SPSS
- Excel
- C
- HTML
- CSS
- Git
- Latex
- Microsoft Office
- (and more)

Academic Information

Publications

"Linear, nested, and quadratic ordered measures: Computation and incorporation into optimization problems"

Blanco, V.; Pozo, Miguel A.; Puerto, J.; Torrejon, A. (2025)
(<https://doi.org/10.48550/arXiv.2503.18097>)



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"A fresh view on Least Quantile of Squares Regression based on new optimization approaches"

Puerto, J.; Torrejon, A. (2025)

Expert System with Applications

(<https://doi.org/10.1016/j.eswa.2025.127705>)

"Benders Decomposition for the Discrete Ordered Median Problem"

Ljubic, I.; Pozo, Miguel A.; Puerto, J.; Torrejon, A. (2024)

European Journal of Operational Research

(<https://doi.org/10.1016/j.ejor.2024.04.030>)

"The Ordered Median Tree Location problem"

Pozo, Miguel A.; Puerto, J.; Torrejon, A. (2024)

Computer and Operations Research

(<https://doi.org/10.1016/j.cor.2024.106746>)

Teaching

2024-2025 (academic year), Statistics

60 hours - Health Engineering Degree, University of Málaga and University of Seville, Seville, Spain.

Jul 2024, IX International Mathematics Summer School

1.5 hours - Faculty of Mathematics, Seville, Spain.

Jul 2024, Advanced Linear Models for Biology

18 hours - Institute of Educational Sciences, Seville, Spain.

Jun 2024, Introduction to R and Linear Models for Biology

10 hours - Institute of Educational Sciences, Seville, Spain.

2023-2024 (academic year), Statistics

60 hours - Health Engineering Degree, University of Málaga and University of Seville, Seville, Spain.

2022-2023 (academic year), Statistics

60 hours - Health Engineering Degree, University of Málaga and University of Seville, Seville, Spain.

Participation in Research Projects

PROYECTO ETNIAS: Strategies for neuroprotective therapy administration for ischemic stroke identified through biomarkers in Seville ambulances

(PI21/01158)

Optimization on data science and network design problems: Large scale network models meet optimization and data science tools

(PID2020-114594GB-C21)

Challenges of combinatorial optimization in new complex network models and data science

(P18-FR-1422)

Nuevos resultados sobre los problemas de diseño y optimización en redes complejas: Aplicaciones al diseño de ciudades inteligentes

(US-1256951)

Communications - Conferences, Workshops and Seminars

"Sorting is all you need"

Nov 7, 2025 - 5th Spanish Young Statisticians and Operational Researchers Meeting, Institute of Mathematics, University of Seville, Seville, Spain.



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"Tras el ictus, antes del deterioro: Pistas neuroprotectoras ocultas en los datos clínicos masivos"

Sep 25, 2025 - Congreso Nacional sobre Innovación mediante el Uso de Datos Sanitarios INNODATA 2025, Seville, Spain.

"The Measure of Everything, a flexible modeling framework for combinatorial problems"

Oct 15–17, 2025 - 5th EUROYoung Workshop, Naples, Italy.

"Order then optimize: A flexible modeling framework in Location"

Sep 17–19, 2025 - 30th Meeting of the EURO Working Group on Locational Analysis & XIV International Workshop on Locational Analysis, University of Cádiz, Jerez, Spain.

"Voyageur, il n'y a pas de chemin... on optimise en marchant"

Jun 16, 2025 - Conference at Colegio de España de París, Paris, France.

"The Measure of Everything, a flexible modeling framework in Location Science"

Jun 13, 2025 - XLI Congreso Nacional de Estadística e Investigación Operativa, University of Lleida, Lleida, Spain.

"Advances in ordered optimization methods"

Jan 15, 2025 - VI Congreso de Jóvenes Investigadores de la Real Sociedad Matemática Española, University of the Basque Country, Bilbao, Spain.

"Revisiting the Least Quantile of Squares Regression Estimator from an Optimization Perspective"

Nov 28, 2024 - Workshop del "Grupo de Trabajo en Transporte de la Sociedad de Estadística e Investigación Operativa 2024", Institute of Mathematics, University of Seville, Seville, Spain.

"Optimización, solvers y R con ejemplos"

Nov 7, 2024 - III Congreso y XIV Jornadas de Usuarios de R, Institute of Mathematics, University of Seville, Seville, Spain.

"Modeling ordered interactions in Location Problems"

Sep 7, 2024 - XIII International Workshop on Locational Analysis and Related Problems, University of Granada, Granada, Spain.

"Ordered interactions in Location Problems"

Jun 19, 2024 - EWGLA XXIX, Sheffield University Management School, Sheffield, United Kingdom.

"Ordered interactions in combinatorial problems"

Jun 7, 2024 - ECCO XXXVII Conference 2024, KU Leuven, Ghent, Belgium.

"Conexiones con precios óptimos"

Jan 22, 2024 - Congreso Bienal de la Real Sociedad Matemática Española, Public University of Navarre, Pamplona, Spain.

"Ordered approach to Equity in Location problems"

Jan 18, 2024 - Workshop on Resource Optimization 2024, University of Cádiz, Cádiz, Spain.

"New Advances in the Stackelberg Minimum Spanning Tree Problem"

Nov 14, 2024 - International Doc Course Closing Workshop, Institute of Mathematics, University of Granada, Granada, Spain.



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"Optimally pricing connections"

Nov 10, 2023 - XL Congreso Nacional de Estadística e Investigación Operativa, Miguel Hernández University of Elche, Elche, Spain.

"Framework for Fairness Analysis in Location Problems"

Sep 8, 2023 - XII International Workshop on Locational Analysis and Related Problems, University of Edinburgh, Edinburgh, United Kingdom.

"Ordered Median Problems and Fairness in Location Theory"

Jun 26, 2023 - ISOLDE XVI and EWGLA XXVIII Conference, Kaiserslautern and Baden-Baden, Germany.

"Concerning fairness in Location Analysis"

Jun 5, 2023 - 3rd EUROYoung, ESSEC Business School, Paris, France

"Improving algorithm efficiency for fairness allocation in optimization"

Feb 7, 2023 - VI Congreso de Jóvenes Investigadores de la Real Sociedad Matemática Española, University of León, León, Spain.

"Recent advances on the Discrete Ordered Median Problem"

Dec 22, 2021 - Seminario "Aplicaciones de la Optimización 2022", Institute of Mathematics, University of Seville, Seville, Spain.

"Ordered median location problems with connection structure"

Nov 10, 2022 - Fourth BYMAT Conference: Bringing Young Mathematicians Together, Polytechnic University of Valencia. Online.

"Optimización, solvers y R con ejemplos"

Sep 28, 2022 - Asociación de Usuarios de R Murcia (UMUR). Online.

"Location, connection and order in discrete networks"

Sep 16, 2022 - EWGLA XXVII, University of Aveiro, Aveiro, Portugal.

"Location, connection and order in discrete network problems"

Jun 22, 2022 - Second EUROYoung Workshop, Faculty of Engineering of Porto, Porto, Portugal.

"The Ordered Median Tree Location Problem"

Jun 8, 2022 - XXXIX Congreso Nacional de Estadística e Investigación Operativa, University of Granada, Granada, Spain.

"Localización, conexión y orden en problemas de redes discretas"

Mar 10, 2022 - Seminario PhD, Institute of Mathematics, University of Seville, Seville, Spain.

"The Ordered Median Tree Location Problem"

Dec 21, 2021 - Seminario "Advances on data analysis, logistics and transportation problems on complex networks", Institute of Mathematics, University of Seville, Seville, Spain.

Communications - Dissemination Activities

"Ganar, ganar y ganar... ¡al billar!"

Nov 11, 2025 - Activity "QUIFIBIOMAT 2025", Faculty of Mathematics, University of Seville, Seville, Spain.

"Matemáticas para la Lengua y la Literatura"

Mar 19, 2025 - I.E.S. La Campiña, Arahal, Seville, Spain.

"Matemáticas para aprobar Lengua y Literatura"

Jan 23, 2025 - Programa IMUS de visitas de institutos de enseñanza secundaria y bachillerato, Institute of Mathematics, University of Seville, Seville, Spain.



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"¿Cómo ser el mejor? Mejor llama a la ordenación"

Dec 13, 2024 - Sixth Edition of Beer and Science, Faculty of Physics, University of Seville, Seville, Spain.

Ponente invitado a la mesa redonda

Dec 5, 2024 - II Jornadas de Jóvenes Investigadores Barreños, City Hall of Los Barrios, Los Barrios, Cádiz, Spain.

"¿Elipses en un billar?"

Nov 11/12, 2024 - Jornadas de Puertas Abiertas "QUIFIBIOMAT 2024", Faculty of Mathematics, University of Seville, Seville, Spain.

"Magia, Potagia, ¡ChatGPT!... ¿o Matemáticas?"

Mar 13, 2024 - Programa de Actividades 2024, Faculty of Mathematics, University of Seville, Seville, Spain.

"Modelos Equitativos"

Nov 24, 2023 - Programa IMUS de visitas de institutos de enseñanza secundaria y bachillerato, Institute of Mathematics, University of Seville, Seville, Spain.

"Matemáticas y Literatura"

Nov 16, 2023 - Jornadas de Puertas Abiertas "QIFIBIOMAT 2023", Faculty of Mathematics, University of Seville, Seville, Spain.

Ponente invitado a la mesa redonda

Oct 30, 2023 - I Jornadas de Jóvenes Investigadores Barreños, City Hall of Los Barrios, Los Barrios, Cádiz, Spain.

"¿Matemáticas y Literatura?"

May 3, 2023 - Curso de divulgación matemática, I.E.S. Carlos Cano, Los Barrios, Cádiz, Spain.

"Un Billar Infalible"

Nov 8, 2021 - Jornadas de Puertas Abiertas "QIFIBIOMAT 2021", Faculty of Mathematics, University of Seville, Seville, Spain.

Management, Committees, Editorial Work and Organization

Technical Editor of *BEIO* - Bulletin of SEIO

Apr 1, 2025 - Present. Society of Statistics, Operations Research and Data Science, Spain.

Member of the Scientific Committee of the *IV Congreso y XV Jornadas de Usuarios de R*

Nov 5–7, 2025. Polytechnic University of Valencia, Valencia, Spain.

Member of the Organizing Committee of the *5th Spanish Young Statisticians and Operational Researchers Meeting (5th SYSORM)*

Nov 5–7, 2025. Institute of Mathematics, University of Seville, Seville, Spain.

Coordinator for dissemination of scientific activities at IMUS

Oct 1, 2023 – Feb 25, 2025. Institute of Mathematics, University of Seville, Seville, Spain.

Organization and promotion tasks at the *30th Meeting of the EURO Working Group on Locational Analysis & XIV International Workshop on Locational Analysis*

Sep 17–19, 2025. University of Cádiz, Jerez, Spain.



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Organizer of the invited session *Localización (GELOCA3)*

Jun 10–13, 2025. XLI Congreso Nacional de Estadística e Investigación Operativa (SEIO 2025), University of Lleida, Lleida, Spain.

Organizer of the session *CD04: Optimización Matemática*

Jan 13–17, 2025. VII Congreso de Jóvenes Investigadores de la Real Sociedad Matemática Española, Bilbao, Spain.

Member of the Organizing Committee of the *III Congreso y XIV Jornadas de Usuarios de R*

Nov 6–8, 2024. Institute of Mathematics, University of Seville, Seville, Spain.

Member of the Organizing Committee of the *XIII International Workshop on Locational Analysis and Related Problems*

Sep 4–6, 2024. Institute of Mathematics, University of Granada, Granada, Spain.

Member of the Organizing Committee of the *3rd Sevilla MINLP Workshop*

Mar 30–31, 2023. Institute of Mathematics, University of Seville, Seville, Spain.

Organizing member of the seminar series *Seminarios PhD*

Academic year 2022–23. Institute of Mathematics, University of Seville, Seville, Spain.

Organizing member of the activity *Los Martes Se Investiga – LMSI*

Academic year 2022–23. Faculty of Mathematics, University of Seville, Seville, Spain.

Achievements

Feb 2016, 2nd Place in Olimpiadas Matemáticas RSME Cádiz

Royal Spanish Mathematical Society
University of Cádiz, Cádiz.

May 2015, Third Prize in Concurso de Retórica y Argumentación

Edith Stein Institute of Philosophy, Granada.

Volunteering

Sep 2021, Science Communicator

University of Seville, Seville
Jornadas de la Noche Europea de los Investigadores.

Academic Years 2015-2016-2017-2020, Collaborating Student

Faculty of Mathematics, University of Seville
Professor: *Jose Antonio Prado Bassas*
Student communicator representing the Faculty of Mathematics

Sep 2012 - Jun 2013, Radio Host

Coordinadora Abril, Los Barrios, Cádiz
Radio program "No me gustan los Lunes" on Radio Sol Los Barrios.